Hackathon kickoff — meeting summary

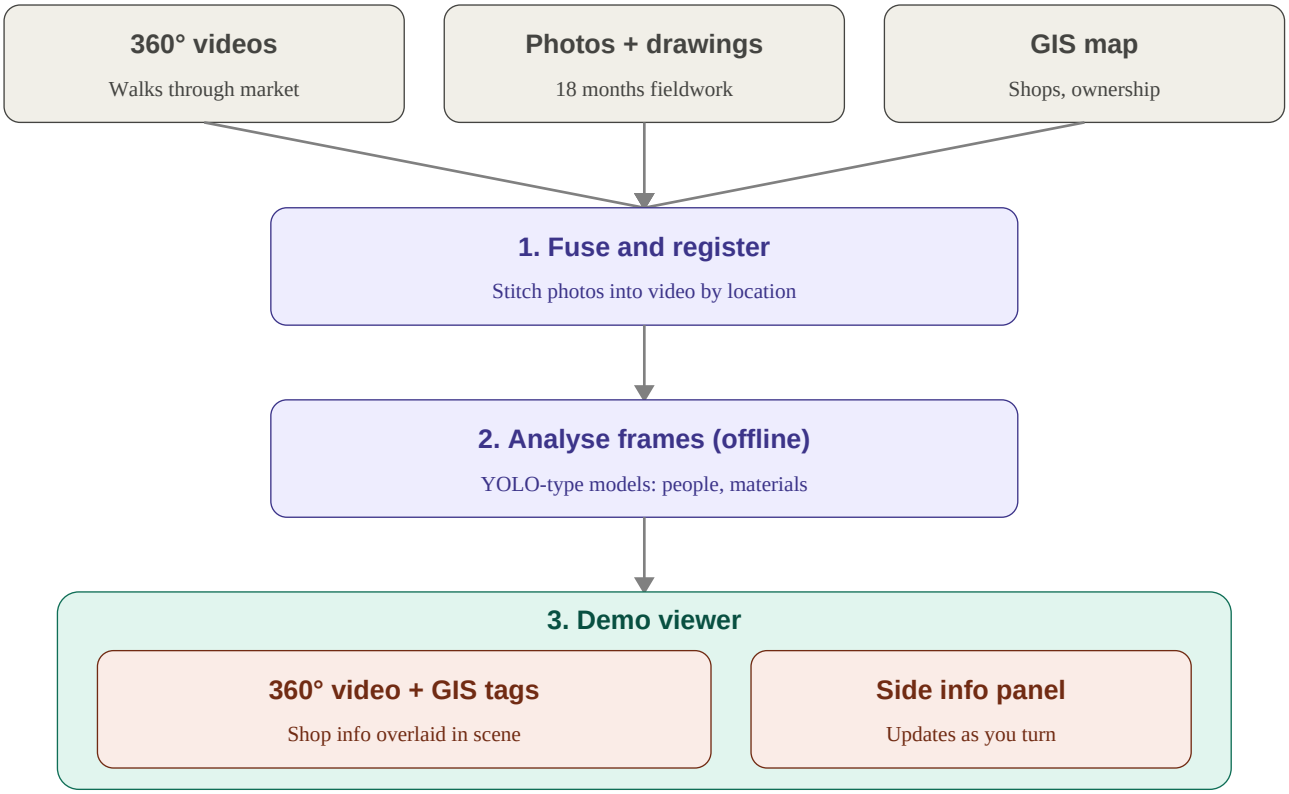
Sunday 11 October 2026 · Participants: Alex, Boaz, Ido

The project

Boaz (PhD, architecture; background in art and anthropology) spent 18 months doing participant observation in Shuk HaPishpeshim, Jaffa, as "the market's artist", collecting drawings, interviews, photos, 360° videos and GIS maps of shops. His long-term goal is a layered digital twin of the market in his illustration style, with layers that can be switched on and off like a building model. A computer-science collaborator is separately working on style transfer and 360° rendering.

Hackathon scope and planned pipeline

Not the full twin. Register photos and GIS data onto the 360° video by location, run off-the-shelf vision models on frames, and show a viewer with GIS shop info overlaid plus a side panel that updates as you rotate (e.g. "64 people, 38% men"; "62% cardboard, 22% metal framing"). Processing is offline; the demo plays back as if live. LLM-based scene editing was set aside as a later extra.



Alex's notes from the meeting

Vision

- **Goal:** take the whole market system (environment) and create it as a digital twin that looks like Boaz's illustrations.
- **Motivation:** the illustrations record emotional information.
- **Method:** 360° spatial video combined with the GIS system; cross-reference all information in one system.

Technical discussion

- Stitching photos together: trained AI models in computer vision to stitch videos together, or videos with photos. One continuous video needs no stitching, but we may still want to stitch photos by their location (option 1). Semantic or old-school methods?
- Initial baseline: scanning the market? Modelling the houses and changing their location? No. The first task may be to stitch all the data we have, then process everything in sequence (building a sort of database).
- Automatically count people; study the data extracted from all scenes. Boaz agreed. This is much simpler than his final vision.
- A tool that scans through videos in real time with GIS, with statistics on people as the video plays (how many men, women?).
- YOLO.
- LLMs can be used for the demo, but as a side feature.
- Initial step: fusion between videos, images and locations.
- A registration model, then YOLO or similar to extract data from the video or images.
- For modifications, multimodal LLMs could switch the houses.
- **Key point: cross-reference the GIS information into the video** (text above it, icon, or different colour).

Who brings what

Alex	Image processing, segmentation, tracking interfaces across frames, custom-trained particle classification
Boaz	Domain knowledge, field data, GIS, visual style
Ido	Robotics MSc (Technion); classical ML and computer vision; proposed stitching/registration + YOLO approa

Action items before Tuesday

- Send the group a summary of agreed tasks (sender not specified in the meeting)
- Set up a shared GitHub repo
- All three sign up for the GPU cluster (link from organisers, shared on WhatsApp)
- Boaz re-sends the data folder link by email
- Everyone reviews the material

Logistics

- Meet in person Tuesday morning
- Alex leaves at 14:00 on Tuesday (job interview); full-time on other days
- No fourth member expected

Open questions to settle early

- Do the 360° videos carry GPS or timestamps? Without them, registration relies on image matching — much harder.
- Materials detection (canvas, cardboard) likely needs a labelled dataset; off-the-shelf YOLO won't cover it.
- Demographic estimates (gender split) carry accuracy and ethics risks — expect judges to ask.